

# MHDRM Cross-Site Readiness Assessment

Prototype screening report for multimodal federated learning

## Assessment Context

|                         |                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------|
| Project name            | AI Breast Cancer Detection                                                                                    |
| MMFL use case           | Development of an AI-enabled screening tool for early breast cancer detection.                                |
| Target outcome or label | Binary patient-level outcome: cancer / no cancer.                                                             |
| Participating sites     | Hospital (NY)<br>Hospital (MA)<br>Hospital (CT)<br>Outpatient oncology center<br>Cancer research institute    |
| Target population       | Adults at risk of breast cancer, with subpopulations grouped by age, demographics, and clinical risk factors. |
| Required modalities     | Unstructured EHR data (e.g., clinical notes)                                                                  |

## Assessment Summary

| Total score | Readiness | Classification                       |
|-------------|-----------|--------------------------------------|
| 12/14       | 85.7%     | Preliminary cross-site compatibility |

## Pillar Results

| Category                 | MHDRM pillar                         | Response                   | Score | Flag |
|--------------------------|--------------------------------------|----------------------------|-------|------|
| Data Characteristics     | Modality Availability                | 1 — Partially demonstrated | 1     |      |
| Data Characteristics     | Data Quality and Completeness        | 2 — Demonstrated           | 2     |      |
| Data Characteristics     | Population Representativeness        | 2 — Demonstrated           | 2     |      |
| Integration Readiness    | Standardization and Interoperability | 1 — Partially demonstrated | 1     |      |
| Integration Readiness    | Cross-Modal Alignment                | 2 — Demonstrated           | 2     |      |
| Organizational Readiness | Privacy and Security by Modality     | 2 — Demonstrated           | 2     |      |
| Organizational Readiness | Provenance and Governance            | 2 — Demonstrated           | 2     |      |

## Evidence and Rationale

| MHDRM pillar                  | Evidence reviewed and scoring rationale                                                                                                                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modality Availability         | Modality-availability requirements are documented. All required modalities are available and confirmed via site audits, but imaging and laboratory coverage vary materially by site based on clinical appropriateness.                        |
| Data Quality and Completeness | Common data-quality dimensions and acceptance thresholds are documented. A standardized outcome variable label has been established. Each site has completed an inventory of their local data and provided aggregated results for evaluation. |

| MHDRM pillar                         | Evidence reviewed and scoring rationale                                                                                                                                                                                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Population Representativeness        | A target population has been defined and documented, along with relevant subgroups. Demographic and clinical distributions were compared across sites, and the consortium documented sampling, weighting, and subgroup-monitoring strategies for identified differences.                              |
| Standardization and Interoperability | Interoperability standards are documented and versioned. Sites use DICOM for imaging and agreed-upon mappings for laboratory codes and units.                                                                                                                                                         |
| Cross-Modal Alignment                | Patient, encounter, and diagnostic-window linkage rules are documented and shared to all sites.                                                                                                                                                                                                       |
| Privacy and Security by Modality     | The consortium completed cross-site privacy and security assessments covering each modality, combined-modality reidentification risks, and federated model updates. All sites implemented compatible controls for deidentification, encryption, access, audit logging, and approved metadata sharing. |
| Provenance and Governance            | The consortium has established data governance protocols that are well documented, with clear ownership and responsibilities.                                                                                                                                                                         |

**Prototype limitation:** This report provides a preliminary preimplementation screening of cross-site data compatibility. The scores and thresholds have not been empirically validated and do not certify that an MMFL initiative will be successful.